

Knowledge Graph Summarization Project Group (KGSUMM PG)

Brainstorming Section

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Recap

Entity Summarization

Entity Summary

► Given:

1. A knowledge graph

$T \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$, i.e., a set of triples (s, p, o) with $s, e \in \mathcal{E}$, $p \in \mathcal{R}$, and $o \in (\mathcal{C} \cup \mathcal{L} \cup \mathcal{E})$.

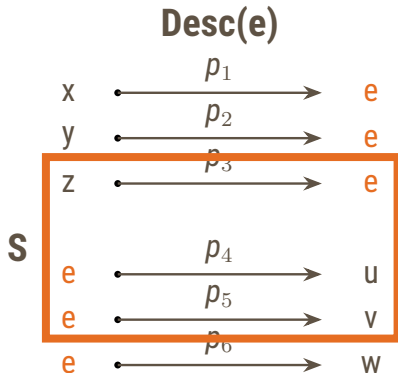
2. An entity description

$\text{Desc}(e, T) \subseteq T$ (e.g., the subset of triples in T that mention an entity e).

$$\text{Desc}(e, T) = \{(e, p, o) \in T \vee (s, p, e) \in T\}$$

► Entity summary (S)

- A summary of an entity e in $\mathcal{T}$ of size k
- Problem definition [1]



$$\underset{S \subseteq \text{Desc}(e)}{\text{find}} \quad \underset{S}{\arg \max} \quad \text{score}(S \mid \mathcal{T}), \quad \text{subject to } |S| \leq k$$

Part 2

Entity Summarization Evaluation Metrics

F-Measure

► Let:

- S = set of triples in the system summary
- $\mathcal{G} = \{G_1, G_2, \dots, G_N\}$ = set of all gold summaries
- G_i = set of triples in the i -th gold summary

For each gold summary $G_i \in \mathcal{G}$:

$$\text{Precision}_i = \frac{|S \cap G_i|}{|S|}, \quad \text{Recall}_i = \frac{|S \cap G_i|}{|G_i|}$$
$$F_1^{(i)} = \begin{cases} \frac{2 \cdot \text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}, & \text{if } |S \cap G_i| > 0, \\ 0, & \text{otherwise.} \end{cases}$$
$$F_{\text{avg}} = \frac{1}{N} \sum_{i=1}^N F_1^{(i)}, \quad N = |\mathcal{G}|$$

- A higher F_{avg} indicates greater overlap and consistency between the system summary and all human-authored gold summaries.

Normalized Discounted Cumulative Gain (NDCG)

Given a ranked list of triples $R = [t_1, t_2, \dots, t_k]$ and a set of gold summaries $\mathcal{G} = \{G_1, G_2, \dots, G_N\}$:

- Each triple t is assigned a relevance grade:

$$rel_t = |\{G_i \in \mathcal{G} \mid t \in G_i\}|$$

- Discounted Cumulative Gain (DCG) at rank k :

$$DCG_k = \sum_{i=1}^k \frac{rel_{t_i}}{\log_2(i+1)}$$

- Ideal DCG (IDCG) using ideally sorted relevance grades:

$$IDCG_k = \sum_{i=1}^k \frac{rel_{t_i}^*}{\log_2(i+1)}$$

- Normalized DCG:

$$NDCG_k = \begin{cases} \frac{DCG_k}{IDCG_k}, & \text{if } IDCG_k > 0 \\ 0, & \text{otherwise} \end{cases}$$

A higher $NDCG_k$ indicates a ranking closer to the ideal order derived from the gold summaries.

Evaluation Metrics

Mean Average Precision (MAP)

Given a ranked list of triples $R = [t_1, t_2, \dots, t_k]$ and a set of gold summaries $\mathcal{G} = \{G_1, G_2, \dots, G_N\}$:

- The Average Precision (AP) for a gold summary G_i :

$$AP(G_i) = \frac{1}{|G_i|} \sum_{j=1}^k P(j) \mathbf{1}[t_j \in G_i]$$

- Precision at rank j :

$$P(j) = \frac{|\{t_m \in R_{1:j} \mid t_m \in G_i\}|}{j}$$

$\mathbf{1}[\cdot]$ = indicator function, 1 if $t_j \in G_i$, 0 otherwise.

- Mean Average Precision across all gold summaries:

$$MAP = \frac{1}{N} \sum_{i=1}^N AP(G_i), \quad N = |\mathcal{G}|$$

A higher *MAP* indicates that relevant triples are ranked earlier, showing stronger agreement between system and gold summaries.

Part 3

Report Findings

Finding Results

Components

- ▶ Performance Analysis
 - ▶ Presents core metrics such as F1-score, MAP, or NDCG.
- ▶ Statistical Test
 - ▶ Commonly used tests: the Wilcoxon signed-rank test (for non-parametric, paired data) and the paired t-test (for normally distributed data).
- ▶ Runtime Analysis
 - ▶ Evaluates the computational efficiency of each model or method.
- ▶ Qualitative evaluation via case study analysis
 - ▶ Select representative samples (e.g., best, worst, or average cases).

An Example of Performance Results and Statistical Test

Table: Average F-measure score based on our model testing via five-fold cross-validation processes to all entities of the benchmark. Statistically significant results ($p < 0.05$) are indicated by ▲. Results that are not significant are marked with ○. Underperform results are denoted with ▼. The first, second and third symbols next to a result in the last row encode the statistical significance of the comparison of GATES with ESA, AutoSUM and DeepLENS, respectively.

	ESBM-DBpedia		ESBM-LMDB		FACES	
	k=5	k=10	k=5	k=10	k=5	k=10
F-Measure						
ESA	0.332	0.532	0.353	0.435	0.153	0.261
AutoSUM	0.372	0.555	0.430	0.520	0.241	0.316
DeepLENS	0.404	0.575	0.469	0.489	0.130	0.248
GATES	0.423 ▲ ○ ○	0.574 ▲ ▲ ▼	0.437 ▲ ○ ▼	0.535 ▲ ▲ ▲	0.254 ▲ ○ ▲	0.324 ▲ ▲ ▲

An Example of Runtime Computation

Table: Comparative Analysis of Training Times and Entity Processing Efficiency among ESLM models. All times are seconds, and the total number of epochs is 50.

Models	Topk	Input Triples	Output Triples	Training Time		Prediction Time
				Total	Mean	Single Triples
DBpedia						
BERT	5	4436	625	329.56	6.59	0.060
BERT + KGE	5	4436	625	404.50	8.09	0.060
ERNIE	5	4436	625	327.39	6.55	0.067
ERNIE + KGE	5	4436	625	328.88	6.58	0.070
T5	5	4436	625	402.96	8.06	0.071
T5 + KGE	5	4436	625	411.86	8.24	0.072
BERT	10	4436	1250	333.62	6.67	0.060
BERT + KGE	10	4436	1250	329.91	6.60	0.059
ERNIE	10	4436	1250	333.32	6.67	0.069
ERNIE + KGE	10	4436	1250	329.24	6.58	0.069
T5	10	4436	1250	403.83	8.07	0.070
T5 + KGE	10	4436	1250	413.38	8.27	0.073

Qualitative Analysis Via a Case Study Analysis

Entity: Ludwigsburg University

Predicate	Object
homepage	http://www.ph-ludwigsburg.de/
almaMater	Ludwigsburg University
type	University
label	Ludwigsburg University
name	Pädagogische ...
type	Organisation
country	Germany
city	Ludwigsburg
subject	Category:Ludwigsburg
type	Educational Institution

→

GATES (F-Measure: 0.600)

Predicate	Object
homepage	http://www.ph-ludwigsburg.de/
label	Ludwigsburg University
name	Pädagogische ...
city	Ludwigsburg
country	Germany

ESLM (F-Measure: 0.666)

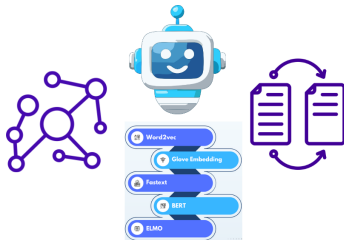
Predicate	Object
country	Germany
city	Ludwigsburg
label	Ludwigsburg University
almaMater	Ludwigsburg University
type	University

Ground Truth Summary

Predicate	Object
almaMater	Ludwigsburg University
type	University
name	Pädagogische ...
city	Ludwigsburg
country	Germany

Figure: ESLM is well aligned with the ground truth summary

That's all Folks!



Thank you!

dice-research.org/teaching/KGSUMM-2025

Have questions?

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All source images are from Google Search and Canva.

- ▶ [1] Liu, Q., Cheng, G., Gunaratna, K. and Qu, Y., 2021. Entity summarization: State of the art and future challenges. Journal of Web Semantics, 69, p.100647.
- ▶ Some content is adopted from <https://sites.google.com/view/entity-summarization-tutorials/www2020>